

MPM Workstreams

Takeaway: These are the core **operational, engineering, logistics, regulatory, and governance workstreams** required to deliver the Mechanical Power Module (MPM) from prototype → pilot → standard → national deployment. Each workstream is structured as a durable operational lane with clear responsibilities, outputs, and cross-dependencies. Guided Links are embedded so you can expand any component.

1. Engineering & Standards Workstreams

These workstreams define the technical backbone of the MPM.

- **Geometry & Interface Standardization** — Owns the mechanical interface, geometry envelope, tolerances, and alignment features.
- **Safety Envelope & Interlocks** — Defines mechanical + electrical interlocks, emergency stop logic, and failure-mode containment.
- **Reference Implementations** — Builds R1 modules, host adapters, latching mechanisms, and compliance prototypes.
- **Measurement & Telemetry Architecture** — Designs torque, speed, thermal, vibration, and interlock telemetry schemas.
- **Compliance Test Suite** — Creates the standardized tests for module, host, and logistics certification.

2. Manufacturing & Fabrication Workstreams

These workstreams ensure the MPM can be produced at pilot and industrial scale.

- **Pilot Module Fabrication** — Produces v0.9/v1.0 modules for pilot deployment.
- **Host-System Adapter Production** — Manufactures adapters for pumps, drives, compressors, and mobile equipment.
- **Material & Wear Analysis** — Evaluates spline wear, latch durability, and long-term interface integrity.
- **Multi-Vendor Manufacturing Enablement** — Ensures competitive neutrality and open access to fabrication.

3. Logistics & Deployment Workstreams

These workstreams treat the MPM as a logistics primitive.

- **Swap-Station Deployment** — Designs, installs, and validates swap stations for rapid module exchange.

- **Transport & Handling Systems** — Builds transport frames, shock-rated carriers, and handling protocols.
- **Operator Training & Certification** — Creates training modules for safe swap-out, inspection, and handling.
- **Pilot Deployment Operations** — Runs controlled pilots across Colorado industrial and municipal sites.

4. Regulatory & Compliance Workstreams

These workstreams align the MPM with federal, state, and standards bodies.

- **DOE / NREL Alignment** — Coordinates CRADA pathways, technical validation, and federal alignment.
- **FERC Permitting Alignment** — Maps MPM deployment to federal permitting modernization.
- **ASME / ISO Standards Harmonization** — Aligns geometry, safety, and logistics with mechanical standards bodies.
- **State & Municipal Compliance** — Ensures Colorado safety, environmental, and industrial compliance.

5. Consortium & Governance Workstreams

These workstreams create the institutional backbone of the MPM.

- **Consortium Formation** — Establishes the Standards Board, Compliance Authority, and Deployment Council.
- **Competitive Neutrality Enforcement** — Ensures open, vendor-agnostic access to the standard.
- **Membership & Onboarding** — Defines membership tiers, onboarding packets, and participation rules.
- **Change-Control & Standards Governance** — Manages revisions to geometry, tolerances, safety envelope, and compliance.

6. Data, Telemetry & Analytics Workstreams

These workstreams turn mechanical performance into standards-grade data.

- **Telemetry Infrastructure** — Builds real-time data pipelines for torque, speed, thermal, vibration, and interlocks.
- **Pilot Data Analysis** — Converts pilot telemetry into geometry, safety, and logistics refinements.
- **Compliance Data Systems** — Creates auditable, regulator-ready compliance logs.
- **Lifecycle & Economic Modeling** — Quantifies cost reductions, uptime improvements, and multi-vendor savings.

7. Deployment & Scaling Workstreams

These workstreams take the MPM from pilot → national → global deployment.

- **Colorado Deployment Program** — Deploys modules across water, industrial, and municipal systems.
- **National Infrastructure Rollout** — Aligns with DOE, FERC, ASME, and state agencies for national adoption.
- **International Standards Harmonization** — Prepares ISO/IEC alignment for global deployment.
- **Manufacturing Scaling** — Enables multi-vendor production at industrial scale.

8. Cross-Workstream Dependencies

These dependencies ensure the roadmap stays coherent:

- Engineering ↔ Manufacturing: tolerances drive fabrication
- Safety ↔ Logistics: interlocks drive swap-station design
- Telemetry ↔ Compliance: data drives certification
- Consortium ↔ Regulatory: governance drives federal alignment
- Deployment ↔ Standards: pilot data drives v1.0 refinements